TEEGALA KRISHNA REDDY ENGINEERING COLLEGE

Department of Artificial Intelligence & Machine Learning

HEAD OF THE DEPARTMENT

Dr. VADIVELAN NATRAJAN

Professor

PROJECT CO-ORDINATOR

Mrs. RANI GADHARI

Assistant Professor

Mrs. TRIVENI

Assistant Professor

AI-Driven Smart Contract Analyzer Integrated with Salesforce Cloud

Project Coordinator :
Mr. P. Ramesh
Assistant Professor

Submitted by:
Indu Priya (22R91A7380)
Vinay Kumar (22R91A7398)
Lakshmi Varshini (22R91A73B7)
Rishi Vardhan (23R95A7310)

Contents

1 Introduction

2 Abstract

3 Existing system

4 Proposed System

5 Requirements

6 Architecture

7 Modules

INTRODUCTION

This project introduces an **AI-driven Smart Contract Analyzer** integrated with **Salesforce Cloud**, designed to streamline and automate the analysis of contracts. By leveraging Natural Language Processing (NLP) and Machine Learning (ML), the system intelligently scans contracts (in PDF/Word format), identifying key clauses such as termination terms, penalties, payment schedules, and data privacy provisions. It then assigns a risk score to the contract, flagging any potential issues or inconsistencies. The analyzer integrates seamlessly with Salesforce, storing the extracted data in structured formats and presenting it in real-time dashboards for easy visualization. Automated alerts notify users of high-risk contracts, and workflows route them to the legal team for further review. Additionally, a Human-in-the-Loop feature allows legal experts to validate or override AI-generated insights, ensuring accuracy and continuous learning. This solution enhances contract compliance, reduces human error, and accelerates decision-making, empowering organizations to manage their contracts more effectively.

ABSTRACT

This project presents an **AI-driven smart contract analyzer** integrated with **Salesforce** to automate contract analysis and streamline legal workflows. The system allows users to upload contracts (in PDF/Word format) into Salesforce, where the AI engine utilizes Natural Language Processing (NLP**)** and Machine Learning (ML) to interpret the contract's text. The AI identifies critical clauses such as termination terms, penalties, payment schedules, and data privacy provisions, while flagging any risky or unusual clauses. Once the contract is analyzed, the system assigns a **risk score** and extracts essential metadata, including dates, parties involved, and renewal terms. This information is then structured and displayed in Salesforce dashboards, giving users a clear view of the contract's compliance status and potential risks. High-risk contracts automatically trigger **alerts**, and workflows are set up to route them to the appropriate legal team for further review. An optional **Human-in-the-Loop** feature enables legal professionals to validate or override the AI’s findings, providing feedback that continuously improves the model's accuracy. This automated process ensures faster, more efficient contract management while reducing human error and improving risk mitigation. The system’s seamless integration with Salesforce makes it a powerful tool for companies seeking to enhance contract compliance and reduce legal risks, all within a unified platform.

EXISTING SYSTEM

In today’s fast-paced business environment, contract management is increasingly complex and time-consuming, especially when dealing with numerous legal agreements across different departments. Traditional methods often lead to inefficiencies, human errors, and missed risks. With the rise of blockchain technology and smart contracts, the need for automated contract analysis has become more pronounced. Many organizations struggle to interpret complex legal clauses and assess potential risks accurately and promptly. To address this, an AI-powered solution integrated with Salesforce Cloud emerges as a powerful tool, enabling businesses to streamline contract management and mitigate legal risks. By combining NLP, ML, and Salesforce, the system can automatically read, analyze, and assess contracts, offering actionable insights in real-time. This project leverages cutting-edge technology to transform how organizations manage contracts, improving accuracy, speed, and compliance across industries.

PROPOSED SYSTEM

The proposed AI-driven Smart Contract Analyzer integrated with Salesforce Cloud offers a comprehensive solution for automating contract analysis and improving contract lifecycle management. This system leverages Natural Language Processing (NLP) and Machine Learning (ML) to automatically read and interpret contracts, whether they are traditional legal documents or blockchain-based smart contracts. By scanning contracts for key clauses—such as termination terms, payment conditions, penalties, data privacy policies, and other critical elements—the analyzer can quickly detect and flag potential risks, inconsistencies, or unusual clauses. The system assigns a risk score to each contract based on its findings, which helps legal teams prioritize contracts for further review. Key contract metadata such as parties involved, renewal terms, deadlines, and dispute resolution clauses are extracted and stored as structured data within Salesforce. This structured data is then presented in real-time dashboards, giving users a clear overview of contract compliance, risk assessment, and contract status. To ensure efficiency, the solution integrates with Salesforce workflows to trigger automatic alerts for high-risk contracts, routing them to the legal team for deeper analysis and validation. The system also includes a Human-in-the-Loop feature, allowing legal experts to verify or override AI-generated insights, further refining the system's accuracy through continuous feedback.

REQUIREMENTS

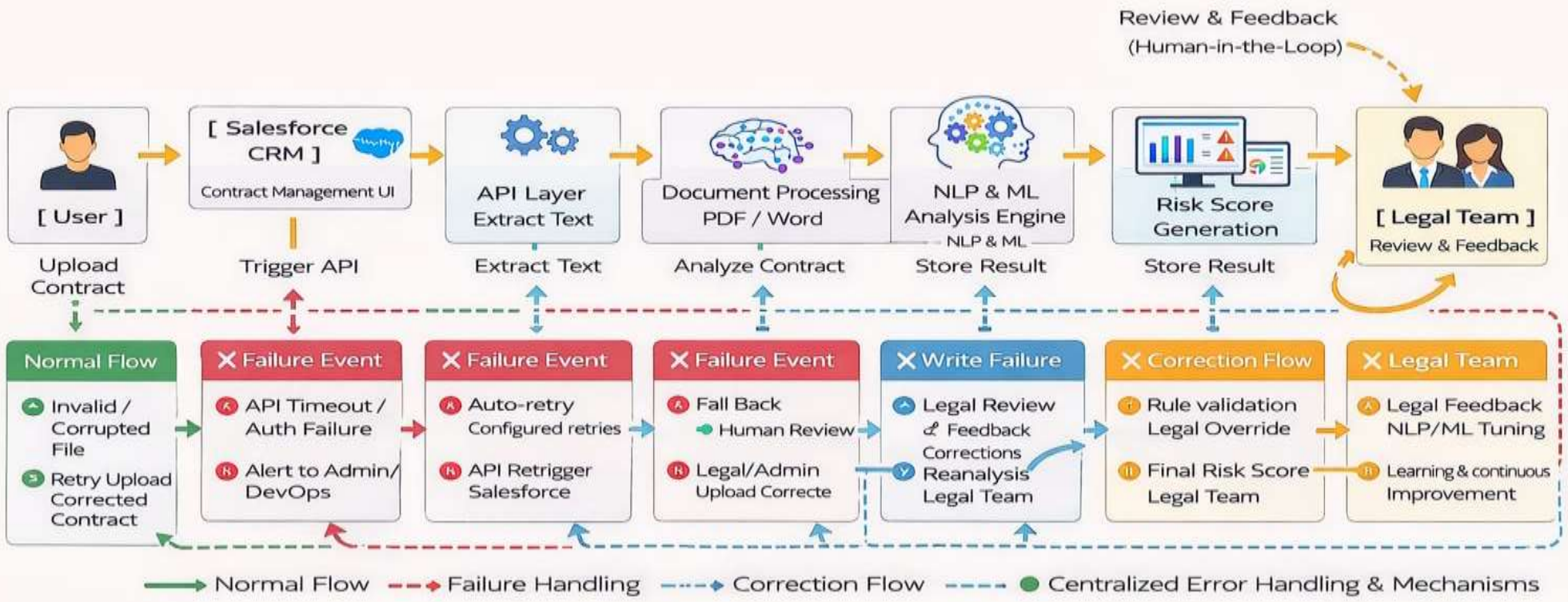
Software

- **Salesforce CRM** (for contract management and AI integration)
- **NLP & ML Libraries** (for contract analysis)
- **Blockchain tools** (for smart contract interactions)
- **Document parsing tools** (for text extraction from PDFs/Word documents)
- **Cloud hosting & storage solutions** (AWS, Azure, etc.)
- **Secure API integrations** (REST APIs, OAuth, etc.)

Hardware

- **Processor**: intel
- **Ram**: minimum 8gb (16 GB recommended for ai tasks)
- **Internet connection**: required for API integration and real-time updates

ARCHITECTURE



MODULES

1. **User Module** – Uploads contracts into the Salesforce system.
2. **Salesforce CRM Module** – Manages contract records and triggers analysis workflows.
3. **API Integration Module** – Securely transfers contract data between Salesforce and AI services.
4. **Document Parsing Module** – Extracts text from PDF and Word contract files.
5. **NLP & ML Analysis Module** – Identifies key clauses and analyze contract risks.
6. **Risk Scoring Module** – Generates a risk score based on contract analysis.
7. **Salesforce Database Module** – Stores contract metadata, risk scores, and analysis results.
8. **Alert & Notification Module** – Sends alerts for high-risk contracts to stakeholders.
9. **Legal Review Module (Human-in-the-Loop)** – Reviews AI results and provides feedback for improvement

THANKYOU